

Functional Stability Theory IV: Cosmological Stability

*A Collector Paper for CRM, Saturation, QG-CRM and Dark Energy under the
FST-UCU Architecture*

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Abstract

This paper is the *cosmological collector slot* (FST-IV) of the Functional Stability Theory (FST) programme, sibling to FST-I (thermodynamic stability of fundamental parameters), FST-II (chemical stability and autocatalytic selection), and FST-III (biological stability and Nash frustration). The cosmological scale of the FST programme is already addressed by several published companion papers: the Cooperative Renormalization Model (CRM) Papers I-IV, the Saturation Theorem (CRM-V), the inflationary QG-CRM line (CRM-VI), and the Dark-Energy variational bound (FST-DE). This collector does *not* replace any of these. It (i) embeds each component in the FST-UCU architecture (universal game, MEPP axiom, three-lemma endgame UCU1-3, RG bridge), (ii) clarifies which UCU step is supplied by which paper, (iii) records the open sub-problems honestly (L4 RG-matching from quadratic gravity to a Hu–Sawicki IR effective theory, the Convexity-Screening No-Go for linear R^2 , and the Pattern-A status of several reductions), and (iv) includes a mathematical reformulation of CRM as a one-parameter Boost subgroup of $\mathrm{PSL}_2(\mathbb{R})$ acting on hyperbolic geometry, opening a Flow-Zeta classification parallel to Riemann (arithmetic), Selberg (geometric-compact), and the non-compact dynamical zoo. **This paper is an honest survey, not a new theorem.** The programmatic umbrella is FST – A Programmatic Hub [1].

Keywords: cosmology, dark energy, cosmological constant, scalar-tensor gravity, quadratic gravity, $f(R)$ gravity, inflation, Saturation Theorem, Pattern A, functional stability theory, FST, Flow-Zeta, hyperbolic geometry.

1 Introduction

The FST programme posits that emergent stability across scales obeys a single conditional architecture: a positive functional, a strict convexity property (UCU1), an existence/coercivity property (UCU2), and a quantitative quadratic lower bound (UCU3) on a closed convex feasibility set $K \subset X$, together with a Renormalization-Group bridge connecting adjacent scales. The application companions FST-I, FST-II and FST-III instantiate this on the particles, chemistry and biology scales respectively. The cosmological scale – which this collector addresses – already hosts a richer set of pre-existing papers, and our task is not to reinvent them but to *place* them in the same UCU/RG language and audit their honest status.

2 The Cosmological Scale at a Glance

Table 1 lists the cosmological FST components.

Component	FST-UCU role	Content	Concept-DOI
CRM Papers I-IV	game-theoretic foundation; UCU1+2 candidate	spieltheoretic CRM, MOND unification, scalaron Lagrangian $R + \gamma R^2$, vector sector	10.5281/zenodo.18728936
CRM-V (Saturation)	<i>Universal</i> tanh normal form	axioms A-D, Abel equation, tanh as unique IR attractor	10.5281/zenodo.19243234
CRM-VI (QG-CRM)	UV completion; L4 RG-matching open	quadratic gravity inflation, Hu-Sawicki IR bridge	10.5281/zenodo.19352449 (draft)
FST-DE	Variational bound; Convexity-Screening No-Go	$\Phi[\rho]$ minimum yields $\Lambda_{\text{eff}} \sim H_0^2 / \ln(M_{\text{Pl}}/H_0)$; gravitational slip $\eta = 5/4$	10.5281/zenodo.19036235

Table 1: Cosmological FST components, their UCU role and persistent identifiers. The collector embeds each in the FST architecture; it does not replace any of them.

3 The Universal Game at Cosmological Scale

TODO v0.2 – v1.0: Identify (x_L, G_L, D_L) for the cosmological scale: x_L is the curvature/density configuration, G_L is the spieltheoretic game from CRM Paper I (player = vacuum field vs. blase), payoff = integrated entropy production over the expansion history. MEPP axiom: cosmological evolution maximises entropy production along open trajectories. RG bridge: connects the $R + \gamma R^2$ scalaron sector to the IR effective $f(R)$.

4 UCU1/UCU2/UCU3 Status across the Cosmological Components

TODO v0.2 – v1.0: Detailed UCU status table. Schematic:

- **UCU1 (strict convexity):** verified for $\Phi[\rho]$ (FST-DE), conditional for the CRM tanh-saturation potential.
- **UCU2 (existence):** Tonelli direct method on reflexive Banach spaces; available throughout the cosmological sector.
- **UCU3 local:** achieved for $\Phi[\rho]$ at the de Sitter fixed point.
- **UCU3 global (L4 RG-matching):** *open*. This is the structural gap CRM-VI documents: lifting the variational bound from $\Phi[\rho]$ through a four-step RG match to an effective $f(R)$ theory at the IR fixed point.

5 Mathematical Reformulation: CRM as Flow-Zeta on Hyperbolic Geometry

TODO v0.2 – v1.0: Integrate the material from the working note CRM_AS_FLOW_ZETA.md. Key structural identifications:

- The CRM composition law $x \oplus y = (x + y)/(1 + xy)$ on $(-1, 1)$ is isomorphic to $(\mathbb{R}, +)$ via $\tanh^{-1}$ and realises a one-parameter Boost subgroup of $\text{PSL}_2(\mathbb{R})$.
- This places CRM in the three-class Zoo taxonomy alongside Riemann (arithmetic), Selberg (geometric-compact) and the non-compact dynamical zeta family (Ruelle-Anosov, BH quasinormal modes).

- The HP-YES claim for CRM is *conditional* on the SGE conjecture (Zoo Master, CORE/zetazoo); it is a classification statement, not an independent proof.

6 Pattern-A Audit

TODO v0.2 – v1.0: Apply the Pattern-A negative form (positivity reformulation alone is in general only a translation of language). The cosmological components contain two clear Pattern-A reductions:

1. **FST-DE Convexity-Screening No-Go:** strict convexity of Φ admits a unique minimiser, so linear $R + \gamma R^2$ cannot produce density-dependent screening. The bridge to Hu–Sawicki is a *conditional structural reduction*, not a closure.
2. **CRM-VI L4 RG-matching:** mapping the QG-CRM UV theory onto a Hu–Sawicki IR $f(R)$ is a sub-conditional reduction with two explicitly identified open sub-conditions (B1 trajectory existence, B2 double-logarithm matching).

7 Honest Status / Open Sub-problems

What this collector delivers.

An FST-UCU embedding of the existing cosmological components (CRM I-IV, CRM-V, CRM-VI, FST-DE); a status map of UCU1/UCU2/UCU3 across them; the Pattern-A audit of the two reductions above; the Flow-Zeta classification of CRM within the Zeta Zoo.

What this collector does NOT deliver.

A closure of the L4 RG-matching (CRM-VI). A non-conditional Convexity-Screening resolution (FST-DE). A non-SGE-conditional HP-YES proof for CRM. A new dark-energy theorem.

Falsification criterion.

If UCU1 or UCU2 fails on the natural cosmological feasibility set, the collector and the underlying components require structural revision. UCU3-global failure is not falsification; it is the locator of the open sub-problem.

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AI Disclosure (Level L3)

This paper was prepared with substantial AI assistance (Anthropic Claude, OpenAI GPT, Google Gemini variants). AI tools were used for cross-paper synthesis between FST-DE and the CRM line, structural editing, status-map generation, Flow-Zeta classification literature cross-referencing, and bilingual EN/DE preparation. All scientific claims, the UCU architecture, the Pattern-A audit and the falsification criterion are the author’s own and are subject to standard authorial responsibility. No section was written autonomously by AI.

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